Alexander DeLise

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Education

Florida State University — Tallahassee, FL
B.S. Computational Science with Honors, GPA: 4.00/4.00 — 2023–2027

- Thesis: Active Learning for Conditional Generative Compressed Sensing
- Advised by Dr. Nick Dexter through the Honors in the Major Program

Florida State University — Tallahassee, FL
B.S. Applied and Computational Mathematics, GPA: 4.00/4.00 — 2023–2027

- Activities: Presidential Scholars Program, Undergraduate Research Opportunity Program, Pi Mu Epsilon Mathematics Honor Society, Society for Industrial and Applied Mathematics

Research Experience

Florida State University — Tallahassee, FL
Honors Thesis — Advised by Dr. Nick Dexter — November 2025 – Present

- Research topics: compressed sensing, generative modeling, inverse problems, active learning, scientific machine learning, representation learning, imaging
- Developed sample complexity theory for conditional generative models under ReLU / Lipschitz assumptions for stable recovery
- Introduced a compatibility factor that quantifies sample complexity and signal recovery penalties for mismatches in conditioning
- Performed extensive image recovery experiments with Stable Diffusion across 15+ NVIDIA A5000 GPUs, showcasing how conditioning affects sampling distributions and recovery
- Languages and tools: Python, PyTorch, Hugging Face, Slurm, Tmux, Bash

Emory University — Atlanta, GA
Computational Math for Data Science REU — Advised by Dr. Matthias Chung — June 2025 – August 2025

- Research topics: scientific machine learning, inverse problems, low-rank matrix approximation, Bayes risk minimization
- Developed a unified Bayes-risk framework and derived closed-form, rank-constrained linear encoder-decoder mappings for forward modeling and inverse recovery
- Generalized optimal mappings to rank-deficient data, operators, and measurements for realistic scientific machine learning settings
- Demonstrated interpretable, competitive baselines in nonlinear problem settings on imaging, finance, and PDE datasets
- Languages and tools: Python, PyTorch

University of Tennessee, Knoxville — Knoxville, TN
Quantum Algorithms and Optimization REU — Advised by Dr. James Ostrowski — May 2024 – April 2025

- Research topics: quantum computing, variational quantum algorithms, quantum optimization
- Developed Partitioned-Constraint QAOA, a hybrid framework that structurally enforces selected constraints through feasible-state preparation and Grover mixing while penalizing the remaining constraints

- Trained reusable variational constraint gadgets for arbitrary low-support constraints and demonstrated improved shallow-depth feasibility and solution quality over penalty-based QAOA
- Languages and tools: Python, PennyLane, PyTorch, Slurm

FAMU-FSU College of Engineering Tallahassee, FL
Operations Research — Advised by Dr. Arda Vanli September 2023 – June 2025

- Research topics: operations research, industrial engineering, mixed-integer nonlinear programming, data-driven healthcare, disease modeling
- Developed a data-driven mixed-integer nonlinear programming model that couples SIRV epidemic dynamics with patient movement
- Tested Florida COVID-19 data to demonstrate how vaccination and mobility jointly affect surge patterns and unmet hospital bed demand
- Languages and tools: Python, Gurobi, GAMS, IBM ILOG CPLEX

Industry Experience

Johns Hopkins University Applied Physics Laboratory Laurel, MD
Data Science Intern May 2026 – August 2026

- **Machine Learning:** Built end-to-end pipeline for training neural operator ML models to predict aerodynamic properties of hypersonic nose-cone geometries, spanning data generation to Slurm-managed model training on CPU cluster
- Led ablation study using a statistical design-of-experiments approach and paired noninferiority testing to evaluate the tradeoff between model size and predictive accuracy across reduced model configurations
- Benchmarked 25+ model variants, identifying reduced configurations that preserved statistically noninferior accuracy while reducing model parameter counts by up to 85% and training time by over 14×
- **Optimization & Decision Science:** Developed a multi-site equipment transfer optimization framework in MATLAB, integrated within a system dynamics logistics model, with an efficient ∼10 second runtime
- Evaluated framework on real data, reducing equipment shortfall by up to 50% and transfers by up to 61%
- Calculated Pareto front of equipment shortfall vs. transfer volume to support readiness and logistics decisions

Publications

Journal Articles

1. **A. DeLise**, K. Loh, K. Patel, M. Teague, A. Arnold, M. Chung. "Optimal Linear Baseline Models for Scientific Machine Learning." *Foundations of Data Science*, 12 (2027), 86–130. [doi] [arXiv]

Conference Proceedings

1. **A. DeLise**, N. Dexter. "Active Learning for Conditional Generative Compressed Sensing." *Advances in Neural Information Processing Systems*, 2026 (accepted). [arXiv]

2. **A. DeLise**, S. Abazari, A. Vanli. "An Epidemiological Mixed-Integer Nonlinear Programming Framework for Vaccine Modeling and Patient Allocation During Pandemics." *International Conference on Industrial Engineering and Operations Management*. [doi]

Preprints

1. A. Wilkie, **A. DeLise**, A. Del Real, R. Herrman, J. Ostrowski. "Partitioned-Constraint QAOA (PC-QAOA): Structural State Preparation and Penalty Enforcement for Quantum Optimization." 2026. [arXiv]

Honors and Awards

- Barry M. Goldwater Scholar 2026
- Pi Mu Epsilon Travel Grant 2025, 2026
- FURC Travel Grant, Florida State University 2025
- Presidential Scholar, Florida State University 2023
- Bright Futures Academic Scholar, State of Florida 2023

Presentations

Contributed Talks

1. Active Learning for Conditional Generative Compressed Sensing [pdf]
   April 2026. ACC Meeting of the Minds, Tallahassee, FL.
2. Optimal Linear Baseline Models for Scientific Machine Learning [pdf]
   January 2026. Joint Mathematics Meetings, AMS Contributed Paper Session on Computer Science, Information, and Communication, Washington, DC.
3. Optimal Linear Baseline Models for Scientific Machine Learning [pdf]
   November 2025. SIAM Student Chapter, Florida State University, Tallahassee, FL.

Poster Presentations

1. Active Learning for Conditional Generative Compressed Sensing [pdf]
   April 2026. 26th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
2. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   March 2026. Florida Undergraduate Research Conference, University of North Florida, Jacksonville, FL.
3. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   January 2026. Joint Mathematics Meetings AMS-PME Student Poster Session, Washington, DC.
4. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   August 2025. CMDS REU Poster Presentation, Emory University, Atlanta, GA.
5. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [pdf]
   April 2025. 25th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
6. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [pdf]
   February 2025. Florida Undergraduate Research Conference, University of South Florida, Tampa, FL.
7. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [pdf]
   January 2025. Joint Mathematics Meetings AMS-PME Student Poster Session, Seattle, WA.
8. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [pdf]
   July 2024. UTK Summer Research Symposium, University of Tennessee, Knoxville, TN.
9. Cost-Effective Location Allocation for COVID-19 Patient Assignment in Florida: A Data-Driven Approach [pdf]
   April 2024. 24th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.

Academic Service

- **President (former Vice President)**, Pi Mu Epsilon, Florida Beta Chapter 2023–Present
  - Lead a community of high-achieving mathematics students, overseeing recruitment, outreach, professional development, and interdisciplinary collaboration while managing chapter finances, communications, and executive coordination.
- **Volunteer Coordinator**, Math Fun Day, FSU Department of Mathematics 2026

  - Organize volunteers and event logistics for hands-on mathematics activities serving K–12 students in Tallahassee.
- **Tutor**, Freelance / Knack Tutoring 2023–2024
  - Provide individualized mathematics and programming support, adapting instruction to diverse learning needs and strengthening problem-solving skills.
- **Founder and Volunteer**, FSU Integration Bee 2024–2025
  - Helped establish a university-wide mathematics tradition, collaborating across organizations to launch the event and later coordinating, volunteering, and competing.
- **Member**, Society for Industrial and Applied Mathematics, Florida State University 2023–Present
  - Engage in professional development and networking through seminars, research presentations, and connections with graduate students and faculty in applied mathematics and computational science.

Press and Media Coverage

1. Florida State University News. "FSU achieves historic milestone: Four students named 2026 Goldwater Scholars." April 2026. [link]